

Engineering Knowledge Request

Information required to build the AI Engineering Platform

07 September 2026

Purpose of this document

The AI engineering platform being built for Vitech turns a customer requirement into a technical specification, a general-arrangement drawing, a bill of materials and a budgetary quotation. Every number it produces is calculated by engineering rules and your own historical designs - the language model only writes the surrounding text. It never invents a dimension, a capacity, a material or a price.

That design has one consequence: the platform can only be as complete as the engineering knowledge it is given. This document lists what we need from Vitech, and explains why each item matters and what it unlocks.

We are not asking for project paperwork. We are asking for the engineering knowledge your application and design engineers already use every day.

How this document is organised

- Part 1 - Engineering assets. The twelve categories of material that teach the platform how Vitech engineers.
- Part 2 - Specific open items. Named gaps that are blocking identified features today. These are the highest-value items in this document.
- Part 3 - Readiness questions. A short set of questions whose answers define how the platform decides a requirement is complete enough to work from.
- Part 4 - Practical notes. Formats, handling and confidentiality.
- Appendix - A checklist you can work through and return.

Partial material is genuinely useful. Please do not hold anything back waiting to assemble a complete set - each item can be put to work on its own, and Part 2 items in particular unblock work immediately.

PART 1 Engineering assets

Listed in priority order. Priority reflects how much each item improves the platform's engineering output, not how hard it is to supply.

Asset	Priority	What it unlocks
1. Customer requirement / enquiry forms	CRITICAL	Defines exactly what your engineers ask a customer, and in what order
2. Design calculation sheets	CRITICAL	Becomes the deterministic calculation engine
3. Engineering rules and design judgement	CRITICAL	The decisions that are not written down anywhere
4. Default assumptions	CRITICAL	Lets the platform proceed sensibly when a customer does not know a value
5. Complete equipment list	HIGH	Defines the scope the platform must cover
6. Standard specifications (completed)	HIGH	Defines the expected output format and content
7. Previous quotations (as issued)	HIGH	Pricing structure, scope, exclusions and commercial language
8. Bills of material	HIGH	How a specification becomes something manufacturable
9. GA drawings	MEDIUM	Components, proportions and drafting conventions
10. Standards followed	MEDIUM	Lets the platform cite the governing standard for a value
11. Document templates	MEDIUM	House format for issued documents
12. Completed project sets	MEDIUM	End-to-end examples for retrieval and validation

1. Customer requirement and enquiry forms

Any form your sales or application engineers use to capture a customer's requirement. Word, Excel, PDF, or a photograph of a printed or handwritten sheet is fine. Superseded and draft versions are welcome.

Why this matters most: these forms record what an experienced Vitech engineer considers necessary before starting work - exactly what the platform needs in order to ask the right questions instead of guessing.

2. Design calculation sheets

Excel calculators, sizing spreadsheets, formula sheets, design books, or the worked longhand calculations an engineer keeps. Please include the constants used and their units.

These become executable engineering. Where a calculation is supplied, the platform computes the value; where it is not, the platform must either copy an older design or report the field as 'To be determined'.

3. Engineering rules and design judgement

The reasoning your engineers apply that is not captured in any calculation sheet. This is usually the hardest material to collect and the most valuable. Examples:

- When a side draft booth is chosen over a down draft booth, and why.
- When fan capacity is increased beyond the calculated figure.
- Preferred materials of construction for specific corrosive duties.
- Standard safety factors and design margins, and when they are varied.
- When a duty is split across multiple machines rather than one larger one.

A recorded conversation or a few bullet points from a senior engineer is perfectly adequate. It does not need to be a formal document.

4. Default assumptions

What your engineers assume when a customer cannot supply a value. Please give the value, and the condition under which it changes. A partial answer is still useful.

A table to complete is provided in the appendix.

5. Complete equipment list

Every equipment type Vitech designs and manufactures, with variants, typical applications, and whether it is still actively sold. Discontinued equipment should be marked as such rather than omitted, so the platform does not offer it.

6. Standard specifications

Completed specifications as issued to customers - ideally 20 to 50 per equipment type, but any number helps. These define which fields appear, in what order, and the wording used.

7. Previous quotations

Quotations as sent to a customer - not the enquiry data sheets already supplied, which are input forms (see Part 2, item B4).

8. Bills of material

Fabrication and purchase BOMs, costed where possible - these show which parts are bought in rather than made.

9. GA drawings

AutoCAD (DWG/DXF) or PDF general-arrangement and layout drawings. DWG or DXF is considerably more useful, because component positions can be read from it rather than estimated.

10. Standards followed

IS, CPCB, OSHA, NFPA, customer-imposed and internal standards - with the clause or value used, where known, so a computed value can cite the standard governing it.

11. Document templates

Specification, quotation and BOM templates, plus inspection reports and test certificates if these are to be generated later.

12. Completed project sets

One complete chain for a real project: the customer's original requirement, the specification issued, the GA drawing, the BOM, the quotation, and any revisions with the reason for each.

Ten such chains are worth more than a hundred loose documents, because they show how one stage led to the next.

PART 2 Specific open items

These are named gaps encountered while building the platform. Each one is currently blocking an identified feature, so these are the highest-value items in this document. Several are small.

Closed since the last request - with thanks

Three items previously on this list have been answered by the workbooks and cost sheet already supplied, and are no longer being asked for:

- Face velocity. Standard Booth.xlsx states 0.5 m/s and builds its whole table from it. The platform now uses 0.5 m/s for the face-based dry booth types, and the value is overridable per design.
- The cropped row on the paint booth cost sheet. It reconciles: MS 18 SWG sheet, 621 kg, material Rs 52,785 + labour Rs 27,945 = Rs 80,730. Added to the visible Rs 5,68,534 this gives Rs 6,49,264 - your stated total, exactly. The booth cost model can now be validated against your own figure.
- Booth panel weight. 27 panels x 23 kg = 621 kg on your worked booth, reproduced exactly, which settled a three-way disagreement in our own model.

One item below - B1, component setting-out - is now the single largest gap between what the platform produces and a drawing Vitech could issue. Everything else in the drawing is finished.

Three further items, B9, B10 and B11, were added after a technical audit of a generated booth drawing. Both sit on the AIRFLOW CHAIN, where one unconfirmed basis changes the blower, the duct and the filter count together - so although they are numbered last, they rank immediately after B1.

B1. Component setting-out rules [HIGHEST PRIORITY]

Where components sit INSIDE the machine: filter bank offset from the rear wall, blower centre-line position, luminaire spacing and mounting height, door and access positions, duct take-off points, nozzle heights, pump position, hopper and airlock centres.

Effect today: this is the ONLY thing between the current output and an issue-quality general arrangement. Overall envelopes are dimensioned correctly from your own formulas; component COUNTS and MODELS are real and come from the resolved specification; the sheets carry sections, hatching, datums and a full item list. But component POSITIONS are drawn indicatively and left undimensioned, and every sheet carries a note saying so, because we will not print a setting-out dimension nobody engineered.

We do not need a document. A marked-up print of one existing GA per equipment type would answer it - or simply the rules an engineer applies from memory. An answer can be as short as this:

"Filter bank sits 150 mm off the rear wall. Blower centre-line is on the booth centre-line, 600 mm above finished floor. Luminaires at 1200 mm centres starting 600 mm from each end wall, mounted at door head height."

Three or four lines of that kind per equipment type converts every component on the drawing from indicative to dimensioned.

B2. Engineering calculations for the two remaining categories

The calculation document supplied covers the Paint Shop Plant, and has been implemented in full. That document states that Powder Coating Plant and Pollution Control Equipment calculations will follow. Those two are still outstanding.

Effect today: for those categories the platform reuses the closest historical design instead of calculating, so a size the archive does not cover is answered less accurately.

B3. Air changes per hour for a drying room or oven

The calculation document does not state one, and unlike face velocity there is no figure elsewhere in the workbooks to take it from.

Effect today: oven exhaust volume is reported as 'To be determined' on every oven specification. Heat load itself is calculated and correct - it follows from mass and temperature rise and needs no ACH - so this single number completes the oven.

B4. A quotation as actually issued to a customer

The three documents received are enquiry and input data sheets, not offers. They have been used for the house document style, which is now applied.

Effect today: the order and content of sections in the generated quotation is our best estimate rather than Vitech's actual format.

B5. Height rule for horizontal baffle wet scrubbers

For vertical spray towers, tower height is computed from the rule supplied. For horizontal baffle units no such rule exists, so the platform cannot establish an overall height and therefore produces no dimensioned drawing for that type.

This is a single rule and it is the only thing blocking that equipment type from drawing.

B6. Reference documents for the knowledge base

Standards, vendor catalogues, design guides, technical literature and training material - in any format.

Effect today: the platform's knowledge search has no documents to search and returns nothing. Technical questions are answered from general engineering knowledge rather than from Vitech's own references.

B7. Thin areas in the historical archive

Thirty-three historical offers are held. Some equipment types are represented by very few records, which limits how well a new requirement can be matched:

Equipment type	On file	Consequence
Water-wash paint booths	1 (vs 13 dry-filter)	A water-wash enquiry is matched against dry-filter designs
Powder coating plants	1	Reuse is effectively a copy of that single plant
Hot air ovens	2 (neither LPG-fired)	An LPG requirement is matched to a diesel-fired design
Booths recording a dry scrubber value	1 (9 sq.m)	The field cannot be filled for a booth of a different size

Additional offers in these areas would improve results immediately, with no software change required.

B8. Rate card and commercial policy

Material rates, fabrication rates, motor and bought-out prices have been supplied and implemented. Still needed: the standard margin policy, and how bought-out items are marked up - bought-out items dominate the cost of a booth, so this is the single largest factor in pricing accuracy.

B9. Which face governs booth airflow [BLOCKS A CUSTOMER-FACING NUMBER]

The platform computes the extract volume as the OPEN FRONT (the booth's length) x an EFFECTIVE FILTER OPENING of 1.5 m x 0.5 m/s x 3600. On your published VT/3.0/DTPB/OP this reproduces 8,100 m³/h exactly, which is why it was adopted.

A second basis is also in circulation, taking the face as roughly 3.0 x 2.5 m. On a 5 m booth the two agree - 5.0 x 1.5 and 3.0 x 2.5 are both 7.5 m², so both give 13,500 m³/h. That agreement is a coincidence of one size, and it hides the disagreement rather than settling it:

Booth length	Open front x 1.5 m	A 3.0 x 2.5 m face
5.0 m	13,500 m ³ /h	13,500 m ³ /h (agree)
6.0 m	16,200 m ³ /h	13,500 m ³ /h (20% apart)
3.0 m	8,100 m ³ /h	13,500 m ³ /h (67% apart)
8.0 m	21,600 m ³ /h	13,500 m ³ /h (60% apart)

Effect today: airflow sizes the blower, the duct diameter and the filter count, so the two bases specify different machines for every booth except a 5 m one. We have implemented the 1.5 m opening because it reproduces your own published figure, and we are not changing it on our own judgement.

Please confirm: which face governs, and whether the 1.5 m effective opening is fixed or varies with booth height or model.

B10. Static pressure basis for blower selection [BLOCKS A CUSTOMER-FACING NUMBER]

Your workbooks carry the rule: never select a blower from CFM alone - use the manufacturer fan curve at the calculated duty point. We can only follow half of it. The platform selects a real catalogue model by AIRFLOW, pinned to the CLP-4 pressure class you build booths around, because no system resistance is calculated anywhere: there is no allowance for filter loading, duct run, bends, plenum or dampers.

Effect today: a selected blower can be shown to deliver the required volume, but NOT that it delivers it against the system it is connected to. On the audited booth the selected CLP-4-10-9000 gives 15,430 m³/h against 13,500 required, and its 89 mmwc is the catalogue figure at the fan's rated point - not a duty point matched on the curve.

Please supply whichever you actually use: a standard static pressure allowance per booth type, or the resistance build-up an engineer works through - and the clean versus dirty filter allowance, since that decides whether the fan still holds duty at the end of a filter's life.

B11. Oven air-circulation basis [BLOCKS A CUSTOMER-FACING NUMBER]

Your heat-load workbook now drives the platform's oven specifications: from a stated size, operating temperature and insulated panel thickness it computes the shell steel mass, the envelope loss and - once a job mass is given - the heating capacity, all from your own cells at your own 30 deg C ambient and 1.2 mm shell.

What it cannot do is size the air. Nothing in the six workbooks states an air-change rate for an oven, a temperature rise across the heater bank, or a fan-selection rule. The heat-load sheet sizes the HEATER and stops there. So the recirculation AIRFLOW is unresolved, and because the blower follows the airflow, the circulation blower HP and quantity are unresolved with it - three rows on every oven specification that we would otherwise be able to fill.

Either input closes all three. Please supply whichever you work from: the air changes per hour you design a batch or conveyerised oven to, or the temperature rise you take across the heater bank. If it differs between a batch oven and a conveyerised one, or with the operating temperature, please say how.

PART 3 Readiness questions

The platform is being extended so that it asks a customer for missing information the way an experienced applications engineer would, rather than producing a draft from an incomplete enquiry. To do that it needs to know what 'enough information' means at each stage of your process.

Key question: for each equipment type, what must always be known before an engineer can confidently prepare (a) a budgetary quotation, (b) a draft specification, and (c) a final release for manufacture?

The distinction matters because these three stages have genuinely different thresholds. A budgetary figure can rest on assumptions that a manufacturing release cannot. A table to complete, one per equipment type, is provided in the appendix.

Three supporting questions

- Which values, if a customer gets them wrong or changes them later, force the design to be redone? These are the ones worth asking about first.
- Which questions can be skipped because the answer can be inferred from the application? For example, does 'for our paint shop' already tell an engineer enough about the contaminant to proceed?
- Which values must never be assumed, and must always be confirmed by the customer in writing before release?

A worked example of what we are asking for

If a customer says only 'we need a wet scrubber for our paint shop', an experienced engineer would not produce a specification. They would ask two or three questions, and assume the rest from experience while stating those assumptions.

We would like to know which two or three questions your engineers would ask first, and which values they would assume rather than ask about. The same answer for each equipment type would let the platform behave the same way.

PART 4 Practical notes

Formats

Native formats are preferred throughout, because they carry information that print formats lose:

Material	Preferred	Also usable
Calculations	Excel with formulas intact	PDF, scan, photograph
Drawings	DWG or DXF	PDF, image
Specifications and quotations	Word or Excel	PDF, scan
BOMs	Excel	PDF, scan
Standards and catalogues	PDF	Scan, printed copy
Engineering rules	Any written note	Recorded conversation, email

An Excel sheet with its formulas intact is far more valuable than the same sheet printed to PDF, because the formula itself is the engineering.

Quantity

More is better, but do not let volume delay a response. One complete project chain, or a single clean calculation sheet, can be put to use immediately. The Part 2 items in particular are mostly small and individually unblock specific features.

Confidentiality and handling

- All material is used solely to build and validate the Vitech platform.
- Nothing is shared with any third party, and nothing is used for any other client.
- Customer names and commercial values may be redacted where you prefer. Engineering content is what matters; a specification with the customer name removed is almost as useful as one with it.
- The platform runs on infrastructure under your control, and customer requirement text is not written into application logs.
- We are happy to sign a non-disclosure agreement before anything is transferred.

What we do not need

- Financial accounts, payroll, or supplier commercial terms.
- Customer contact details or correspondence.
- Anything under a customer confidentiality obligation you cannot discharge.

Suggested next step

A short working session with one senior application engineer and one design engineer would likely produce more than a long document exchange - particularly for Part 1 item 3 and Part 3, where the knowledge is experience rather than paperwork. One hour with the right person is often enough.

APPENDIX A Checklist

For working through and returning. Partial is fine - please send items as they become available rather than holding a complete set.

Part 1 - Engineering assets

- ☐ Complete equipment list, with variants and current sales status
- ☐ Customer requirement / enquiry forms (any format, including superseded versions)
- ☐ Design calculation sheets (Excel with formulas preferred)
- ☐ Engineering rules and design judgement (written notes or a recorded discussion)
- ☐ Default assumptions table (Part 1, item 4)
- ☐ Standard specifications, per equipment type
- ☐ Quotations as issued to customers
- ☐ Bills of material, costed where possible
- ☐ GA drawings (DWG or DXF preferred)
- ☐ Standards followed, with clause or value where known
- ☐ Document templates
- ☐ Completed project sets: requirement, specification, drawing, BOM, quotation, revisions

Part 2 - Specific open items

- ☐ B1. Component setting-out rules - HIGHEST PRIORITY, see the worked example
- ☐ B2. Powder Coating Plant calculations
- ☐ B2. Pollution Control Equipment calculations
- ☐ B3. Air changes per hour for drying room / oven
- ☐ B4. One quotation as actually issued to a customer
- ☐ B5. Height rule for horizontal baffle wet scrubbers
- ☐ B6. Reference documents for the knowledge base
- ☐ B7. Additional offers: water-wash booths, powder coating plants, ovens (incl. LPG)
- ☐ B8. Margin policy and bought-out mark-up
- ☐ B9. Which face governs booth airflow - BLOCKS A CUSTOMER-FACING NUMBER
- ☐ B10. Static pressure basis for blower selection - BLOCKS A CUSTOMER-FACING NUMBER
- ☐ B11. Oven air-circulation basis - BLOCKS A CUSTOMER-FACING NUMBER

Part 3 - Readiness questions

- ☐ Required information per equipment, per stage (budgetary / draft / release)
- ☐ Values that force a redesign if they change
- ☐ Questions that can be skipped because the answer is inferable
- ☐ Values that must always be confirmed in writing before release

APPENDIX B Forms to complete

B-1. Default assumptions

The value your engineers assume when the customer cannot supply one, and what would make them change it. Partial answers are welcome - please leave a row blank rather than guessing.

Parameter	Unit	Default assumed	When does it change
Face velocity (per booth type)	m/s		
Air changes per hour (oven / drying room)	ACH		
Ambient / inlet temperature	deg C		
Transfer efficiency / overspray	%		
Design safety factor	-		
Removal efficiency (scrubber)	%		
Filter media velocity	m/s		
Duct transport velocity	m/s		
Electrical supply	V/ph/Hz		
Installation (indoor / outdoor)	-		

B-2. Required information by stage

One table per equipment type. For each item of information, mark whether it is REQUIRED, may be ASSUMED, or is NOT NEEDED at that stage.

Equipment type:

Information required from the customer	Budgetary quotation	Draft specification	Manufacturing release

Return to

[Contact name] | [email] | [phone]

Please raise any question about a specific item rather than omitting it - in most cases a partial or informal answer is still directly usable.